

Executive Summary –

Customer Churn Analysis (EDA)

1. Overall Churn Performance

- The analysis is based on **7,043 telecom customers**.
 - **1,869 customers have churned**, resulting in an overall churn rate of **26.54%**.
 - **5,174 customers (73.46%)** have remained with the company.
 - This means that approximately **1 out of every 4 customers** has discontinued the service.
 - The results indicate that customer churn is influenced by multiple factors, particularly **tenure, contract type, senior-citizen status, internet service, additional services, and payment method**.
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2. Key Customer Demographic Insights

Gender

- Male and female customers show **relatively similar churn patterns**.
- Therefore, **gender does not appear to be a strong factor** for predicting customer churn.
- Retention strategies should focus more on behavioral and service-related characteristics rather than gender.

Senior Citizens

- Senior citizens show a **significantly higher churn rate** compared with non-senior customers.
 - **Senior Citizen Churn: ~41.7%**
 - **Non-Senior Citizen Churn: ~23.6%**
 - This represents a difference of approximately **18 percentage points**.
 - Senior citizens should therefore be considered a **high-priority customer segment** for retention activities.
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3. Tenure – High Risk During the Early Customer Journey

- Customer tenure is one of the **most important patterns** identified in the analysis.
- Customers who have stayed with the company for a longer period generally show **better retention**.
- Customers in the early stage of their relationship with the company are much more likely to leave.
- Customers with only **1–2 months of tenure show approximately 40% churn**.
- This highlights the importance of the **first few months of the customer journey**.

Business Implication

- The company should focus strongly on **new-customer onboarding and early engagement**.
 - Regular follow-ups, service-quality checks, technical assistance, and clear communication can help reduce early-stage churn.
 - Converting new customers into long-term customers should be a major retention objective.
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4. Contract Type – A Major Retention Factor

- Contract type shows a strong relationship with customer churn.
- The analysis indicates that customers on **month-to-month plans account for approximately 70% of the churned customer group**.
- Customers with **longer-term contracts** show significantly better retention.
- Month-to-month customers may have a higher tendency to switch because they have greater flexibility to leave.

Business Implication

- The company should encourage customers to move toward **longer-term contracts**.
- Possible strategies include:
 - Contract upgrade offers
 - Loyalty benefits
 - Discounts for longer commitments

- Bundled services
- Special benefits for customers shifting from monthly to yearly plans

Important: The ~70% figure represents the share of the churned group associated with month-to-month plans; it should not be interpreted as saying that 70% of all month-to-month customers churn.

5. Internet Service – Fiber Customers Show Higher Risk

Internet service type shows one of the **clearest differences in churn**.

- **Fiber Optic churn: ~40%**
- **DSL churn: ~20%**
- **No Internet Service: <10%**

Key Insight

- Fiber optic customers show **approximately twice the churn rate of DSL customers**.
 - This makes fiber optic customers an important segment for further investigation.
 - The company should examine whether the higher churn is related to:
 - Service expectations
 - Pricing
 - Technical issues
 - Customer support
 - Perceived value
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6. Additional Services Can Improve Retention

The analysis shows an important relationship between additional services and customer churn.

Customers **without** services such as:

- **Online Security**
- **Online Backup**
- **Device Protection**

- **Tech Support**

show churn of approximately **40%**.

In comparison, customers using these services generally show churn of around **15–25%**, depending on the service.

Key Insight

- Customers using additional support and protection services appear to have **better retention**.
- These services may increase the overall value customers receive from the company.
- A particularly important high-risk segment is **fiber optic customers without security or support-related services**.

Business Opportunity

- Promote additional services through:
 - Bundle offers
 - Personalized recommendations
 - Discounts for combined plans
 - Free trials for selected services
 - Targeted offers for high-risk customers
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7. Streaming, Phone Service and Multiple Lines

Streaming Services

- Streaming TV and Streaming Movies show relatively similar churn patterns.
- Churn is approximately **30%** across the observed groups.
- Therefore, streaming services appear to be **weaker churn indicators** compared with contract type, tenure, or internet service.

Phone Service and Multiple Lines

- Phone Service and Multiple Lines also show relatively similar churn patterns.
- These variables do not appear to be among the **strongest drivers of customer churn**.

- Therefore, they should have lower priority in retention strategies compared with the major risk factors identified above.
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8. Payment Method

- The analysis identifies **Electronic Check** customers as having higher churn.
- This indicates that payment method may provide an additional signal for identifying customers at risk.
- However, the notebook does **not provide a specific churn percentage** for Electronic Check customers, so an exact percentage should not be assigned to this finding.

Business Implication

- Electronic Check customers can be included in **churn-risk monitoring**.
 - The company could encourage convenient alternative payment methods through suitable communication or incentives.
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9. High-Risk Customer Profile

Based on the analysis, the following characteristics are associated with **higher churn risk**:

- **Senior citizen**
- **Very low tenure / new customer**
- **Month-to-month contract**
- **Fiber optic internet service**
- **No Online Security**
- **No Tech Support**
- **No Online Backup**
- **No Device Protection**
- **Electronic Check payment method**

A customer having several of these characteristics simultaneously could represent a **high-priority retention opportunity**.

10. Key Business Recommendations

1. Focus on New Customers

- Give special attention to customers during their **first 1–2 months**.
- Improve onboarding and early customer support.
- Conduct proactive satisfaction checks.

2. Promote Long-Term Contracts

- Encourage month-to-month customers to move toward **yearly or longer-term plans**.
- Use loyalty rewards, discounts, and attractive bundled offers.

3. Target Senior Citizens

- Develop dedicated retention strategies for the **41.7% churn segment**.
- Provide simple communication, reliable support, and suitable service packages.

4. Investigate Fiber Optic Churn

- The approximately **40% churn rate** makes fiber customers a major area of concern.
- Review service quality, pricing, technical issues, and customer support.

5. Promote Additional Services

- Encourage customers to adopt **Online Security, Tech Support, Online Backup, and Device Protection**.
- Bundle these services with internet plans to increase perceived value and improve retention.

6. Monitor Payment Behavior

- Keep **Electronic Check customers** under closer churn-risk monitoring.
- Explore whether alternative payment options can improve customer retention.

7. Build Customer Risk Segments

- Instead of treating all customers equally, use the identified factors to create **low-, medium-, and high-risk customer segments**.
- Prioritize retention campaigns toward customers showing multiple risk characteristics.

11. Final Conclusion

- The analysis shows an overall customer churn rate of **26.54%**, which represents a significant retention challenge.
- The strongest churn patterns are associated with:
 - **Early customer tenure**
 - **Month-to-month contracts**
 - **Senior-citizen customers**
 - **Fiber optic internet service**
 - **Lack of additional support and protection services**
 - **Electronic Check payment**
- At the same time, **gender, streaming services, Phone Service, and Multiple Lines show relatively weaker differences in churn.**
- The most important business takeaway is that churn should be addressed through a **targeted customer-retention strategy rather than a one-size-fits-all approach.**
- By focusing on **new customers, high-risk senior customers, month-to-month users, fiber optic customers, and customers without value-added services**, the company can identify customers who need attention before they leave.

Overall Insight

The analysis suggests that improving the early customer experience, encouraging long-term contracts, strengthening fiber customer satisfaction, and increasing adoption of support and protection services can play an important role in reducing customer churn and improving long-term customer retention.